

# Representational Geometry in Neural Networks

From Vision Transformers to RLHF-Aligned Language Models

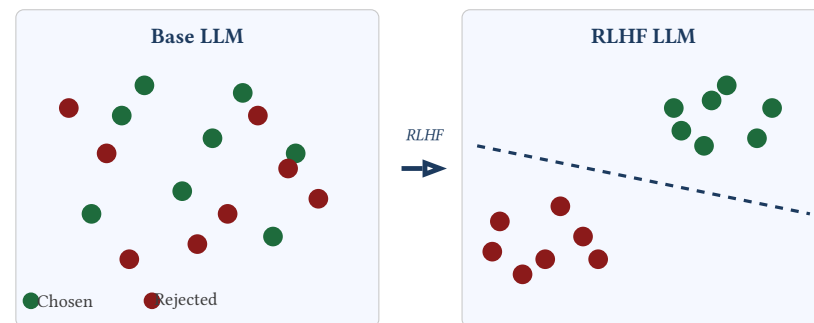
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# Motivation

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# Why Study Representational Geometry?

- Neural networks are powerful — but **what do their internal representations look like?**
- Layers encode rich geometric structure: clusters, manifolds, linear subspaces
- Training transforms this geometry in non-trivial, measurable ways

**Goal:** Perform a geometric audit of what training does to the internal state of a neural network — making the “blackbox” of embeddings literally visible and explorable.

# Research Questions

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# What We Are Investigating

| Part     | Question                                                                                                                      | Method                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Vision   | How similar are CNN vs. ViT internal representations, and which image regions drive cluster formation?                        | CKA + Class Activation Maps    |
| Language | Can a linear hyperplane separate preferred from rejected response vectors after RLHF, and how does this evolve across layers? | LinearSVC + Margin Analysis    |
| Language | Can political or cultural biases introduced by RLHF be identified as geometric structures?                                    | Bias Probes + Cluster Analysis |

# Part 1 — Vision Models

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Two dominant paradigms for image recognition, trained on **CIFAR-10**:

## CNN

- Inductive bias: local spatial structure
- Hierarchical feature extraction
- Fixed receptive field per layer

## Vision Transformer (ViT)

- Global self-attention from layer 1
- Patch-based tokenization
- No explicit spatial inductive bias

**Question:** Do they encode the same information in their intermediate layers, despite fundamentally different architectures?

**Definition: CKA** Given activation matrices  $X \in \mathbb{R}^{n \times p}$  and  $Y \in \mathbb{R}^{n \times q}$ , **linear CKA** is:

$$\text{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \cdot \|Y^\top Y\|_F}$$

$\text{CKA} \in [0, 1]$ , invariant to orthogonal transforms and isotropic scaling.

- $\text{CKA} = 1$ : representations are linearly equivalent
- $\text{CKA} = 0$ : representations are unrelated
- Enables **layer-by-layer** comparison across different architectures



**CAMs** reveal which spatial regions of an image drive a model's prediction:

$$\text{CAM}_c(x, y) = \sum_k w_k^c \cdot A_{k(x, y)}$$

where  $A_k$  is the  $k$ -th feature map and  $w_k^c$  are class-specific weights.

**Remark** Overlaying CAMs on images and comparing CNN vs. ViT lets us ask: do architectures with high CKA also attend to the **same spatial regions**?

## Part 2 — Language Models

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**Reinforcement Learning from Human Feedback** fine-tunes a language model to produce outputs humans prefer.

## **Base: Llama-3-8B**

Pretrained on raw text

No explicit preference signal

Embedding space reflects language structure

## **Aligned: Llama-3-8B-Instruct**

RLHF fine-tuned

Optimized for human preference

Embedding space geometrically reorganized

# Linear Separability of Preferences

**Key question:** Can a linear hyperplane separate preferred from rejected response vectors?

**Insight: Linear Probe** We train a **LinearSVC** on embeddings of chosen vs. rejected responses from Anthropic/hh-rlhf. High accuracy  $\Rightarrow$  preferences are **linearly encoded** in the geometry.

- Evaluated per-layer across all transformer layers
- Margin width as proxy for **geometric separation strength**
- Base vs. aligned model comparison quantifies the alignment effect

RLHF may introduce systematic biases that manifest as geometric structure:

- **Political bias:** Do “left-leaning” vs. “right-leaning” prompts cluster separately?
- **Cultural bias:** Are certain demographic groups encoded into separable subspaces?

**Remark** If biases are **linearly encoded**, they can in principle be identified and mitigated via geometric interventions — e.g. activation steering or representation editing.

# Datasets & Methods

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| Part     | Dataset              | Methods                       |
|----------|----------------------|-------------------------------|
| Vision   | CIFAR-10             | CKA, Class Activation Maps    |
| Language | Anthropic/hh-rlhf    | LinearSVC, Margin Analysis    |
| Language | OpenAssistant/oasst1 | Bias Probes, Cluster Analysis |

**Toolkit:** PyTorch · HuggingFace Transformers · TruncatedSVD · UMAP · t-SNE · Plotly Dash

# Implementation

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A unified **Plotly Dash** app (`app/app.py`) with two tabs — one per research part.

## Tab 1 — CNN vs ViT

- CKA heatmap: pairwise layer similarity
- Side-by-side CAM comparison
- Class filter dropdown

## Tab 2 — RLHF Embeddings

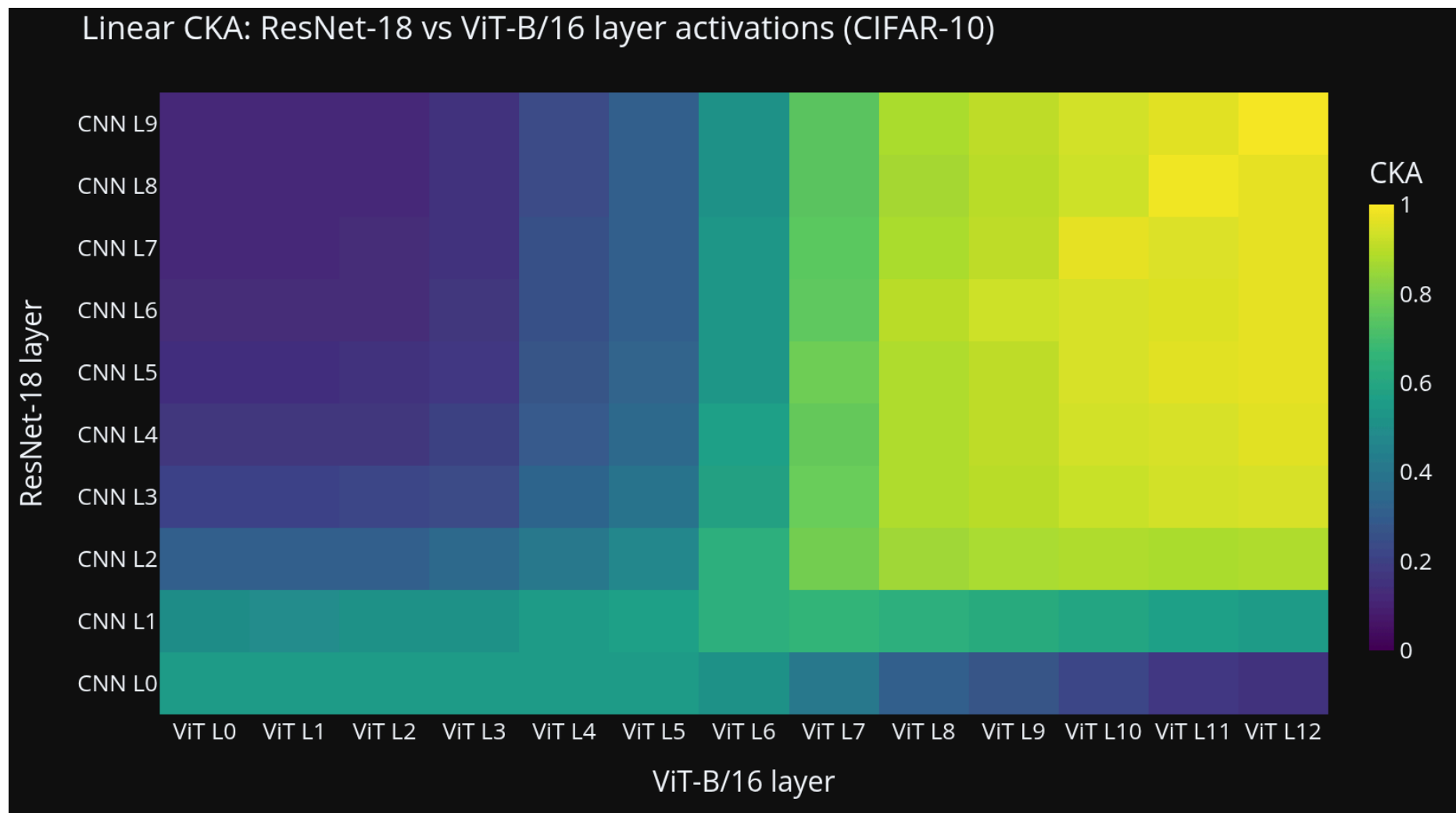
- UMAP / t-SNE scatter per layer
- Layer slider + model toggle
- LinearSVC score line chart

**Offline pipeline:** extraction scripts → HDF5 → cached compute → dashboard

**Mock data generators** allow full UI testing without GPU or model access.

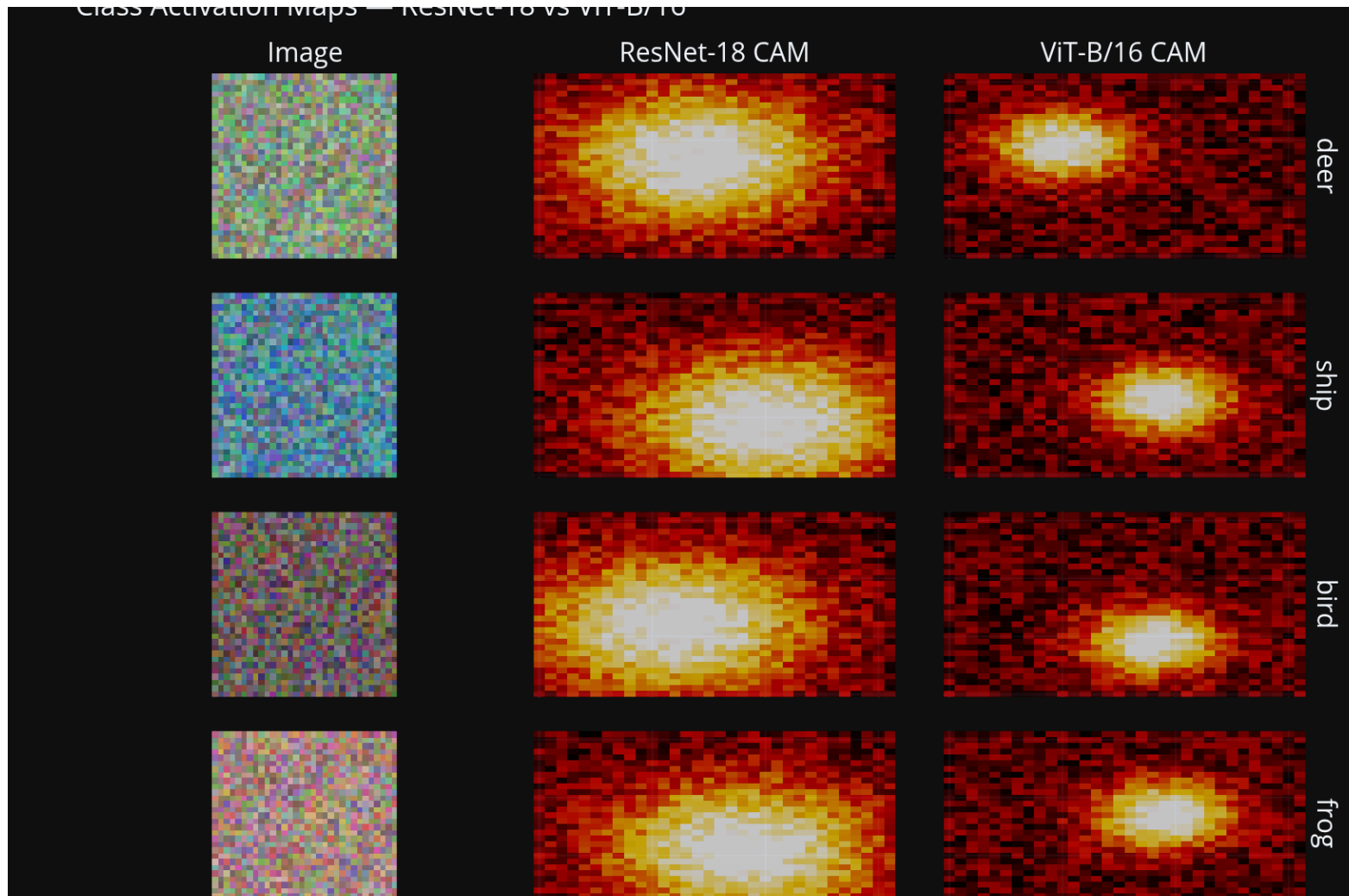


# Part 1 Dashboard — CKA Heatmap



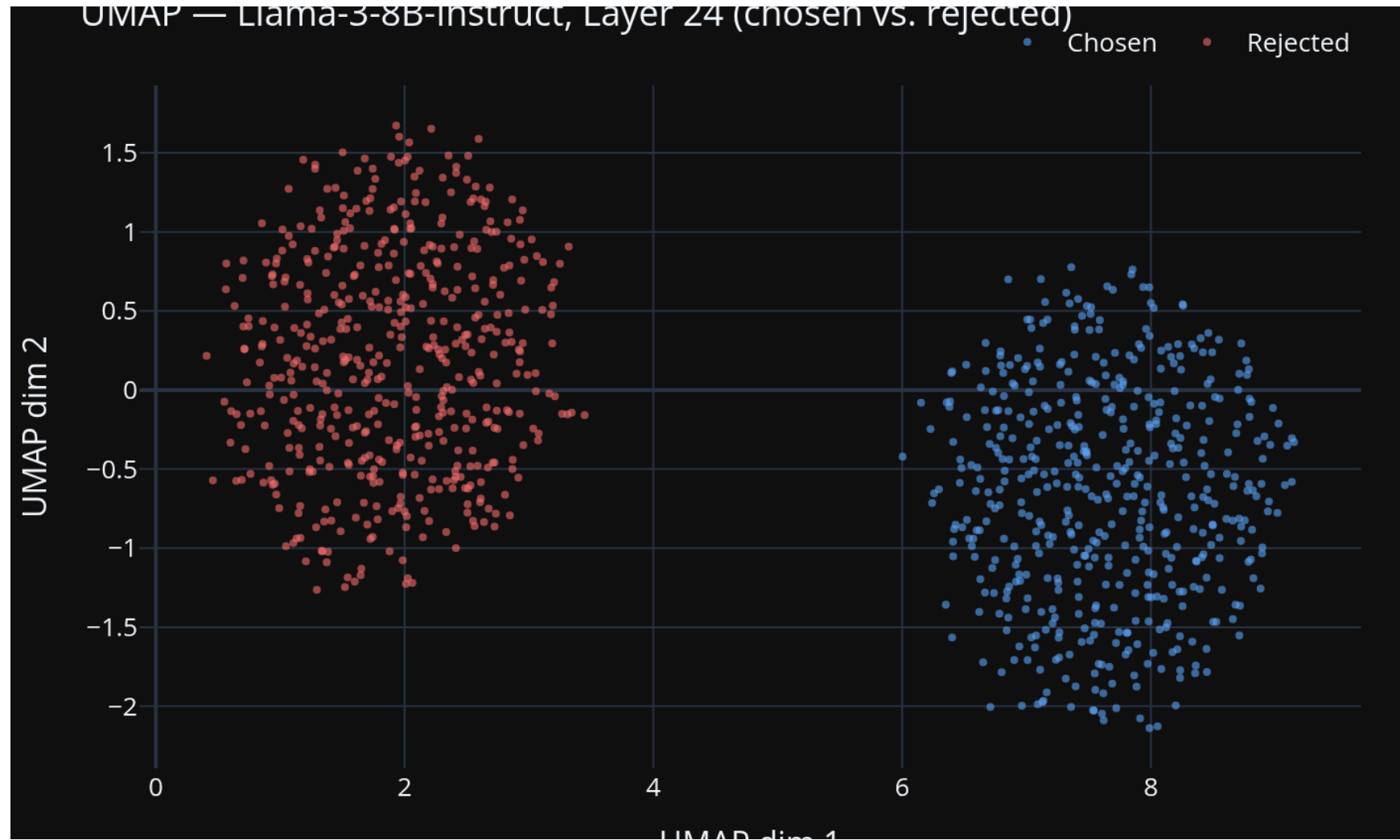


# Part 1 Dashboard — Class Activation Maps

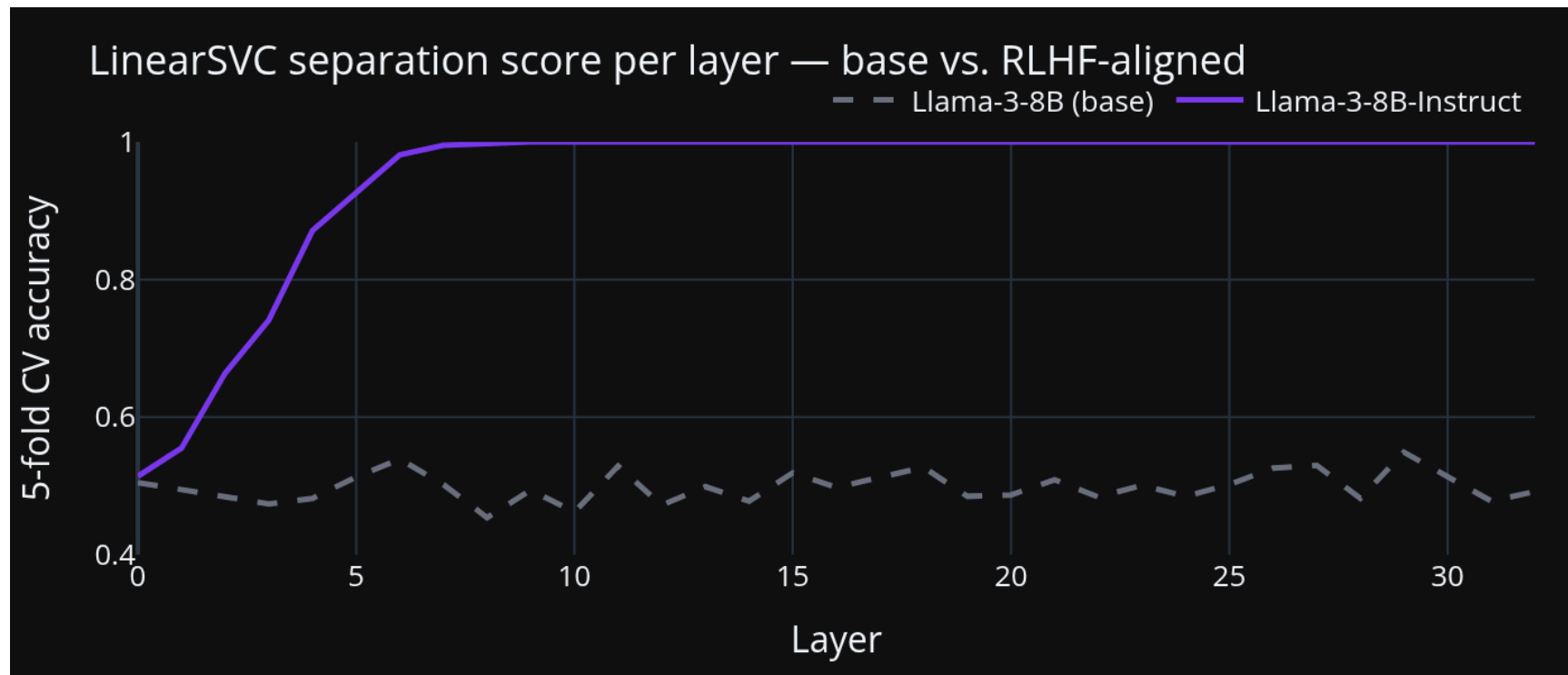




## Part 2 Dashboard — RLHF Embedding Space



## Part 2 Dashboard — LinearSVC Separation Score





**Insight: RLHF Effect** The aligned model shows markedly higher separation accuracy in deeper layers — RLHF geometrically reorganizes the embedding space to linearly encode human preference.

# Summary

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| Part     | Question                                 | Expected Finding                            |
|----------|------------------------------------------|---------------------------------------------|
| Vision   | CNN vs. ViT representational similarity? | CKA profile diverges in mid-layers          |
| Vision   | Which regions drive clusters?            | CAMs reveal architecture-specific attention |
| Language | Linear separability after RLHF?          | High LinearSVC accuracy post-alignment      |
| Language | Bias as geometric structure?             | Cluster analysis reveals bias subspaces     |

## What we aim to show:

- RLHF geometrically reorganizes embedding space in a measurable, linear way
- CNN and ViT representations diverge in intermediate layers despite similar accuracy
- Biases introduced by RLHF are geometrically identifiable via linear probes

## Open questions:

- Does geometric separation correlate with downstream task performance?
- Which layers are most affected by RLHF — early, middle, or late?
- Are the discovered bias subspaces **causally** linked to model outputs?

# References

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